

→ 4 Stroke Engine

4 stroke in each cylinder, 2 revolutions of crank shaft

→ Ideal gas

⊗ Eqn of state: $PV = RT$
 $PV = MRT$

⊗ $\Delta h, \Delta u$: $U(T_2) - U(T_1) = \int_{T_1}^{T_2} C_v(T) dT$

$h(T_2) - h(T_1) = \int_{T_1}^{T_2} C_p(T) dT$

⊗ Constant specific heats

$\Delta u = U(T_2) - U(T_1) = C_v(T_2 - T_1)$

$\Delta h = h(T_2) - h(T_1) = C_p(T_2 - T_1)$

Table: Ideal gas specific heats

Variable specific heats

$U(T)$ & $h(T)$ from tables:
 Ideal gas properties of air.

⊗ ΔS / change in Entropy

$S(T_2, V_2) - S(T_1, V_1) = \int_{T_1}^{T_2} C_v(T) \frac{dT}{T} + R \ln \frac{V_2}{V_1}$

Constant specific heats: $= C_v \ln \frac{T_2}{T_1} + R \ln \frac{V_2}{V_1}$

$S(T_2, P_2) - S(T_1, P_1) = C_p \ln \frac{T_2}{T_1} - R \ln \frac{P_2}{P_1}$

⊗ $\Delta S = 0$: Equal sp. entropy / Isentropic

$\frac{T_2}{T_1} = \left(\frac{P_2}{P_1} \right)^{(k-1)/k}$

$\frac{T_2}{T_1} = \left(\frac{V_1}{V_2} \right)^{k-1}$

$\frac{P_2}{P_1} = \left(\frac{V_1}{V_2} \right)^k$

$k = \frac{C_p}{C_v}$; from table

Variable specific heats

$S(T_2, P_2) - S(T_1, P_1) = S^\circ(T_2) - S^\circ(T_1) - R \ln \frac{P_2}{P_1}$

$S^\circ(T)$ from tables
 A-17

Variable sp. heats - air only

$\frac{P_2}{P_1} = \frac{P_{r2}}{P_{r1}}$

$\frac{V_2}{V_1} = \frac{V_{r2}}{V_{r1}}$

Air only
 P_r, V_r from
 table of air
 A-17

Oil & Time $\rightarrow$ Production will be costly

$\rightarrow$ Fuel efficient car

- Light weight, high -
- Strength materials,
- Automatic shuts, when car is idle

For isentropic process :

Air table

$$V_{r2} = V_{r1} \left(\frac{V_2}{V_1} \right) = \frac{V_{r1}}{r}$$

$$V_{r4} = V_{r3} \left(\frac{V_4}{V_3} \right) = r V_{r3}$$

Since: $V_3 = V_2$ & $V_4 = V_1$, $r = \frac{V_1}{V_2} = \frac{V_4}{V_3}$

V_r from table using Temperature

Otto cycle using cold air-standard basis:
 $T = 25^\circ\text{C}$, R_p , $C_v = C$

Isentropic: $\frac{T_2}{T_1} = \left(\frac{V_1}{V_2} \right)^{k-1} = r^{k-1} \quad (k=C)$

$$\frac{T_4}{T_3} = \left(\frac{V_3}{V_4} \right)^{k-1} = \frac{1}{r^{k-1}} \quad (k=C)$$

$$k = C_p / C_v$$

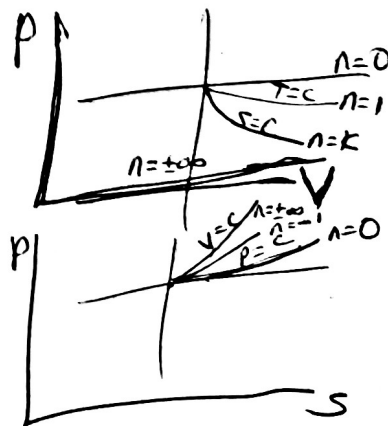
$PV^n = C \rightarrow$ Polytropic process

$PV^k = C \rightarrow$ Polytropic, Isentropic, ideal gas

$PV^n = C$: $n=1 \Rightarrow$ Isothermal ($T=C$)

$n=0 \Rightarrow$ Isobaric ($P=C$)

$n=\pm\infty \Rightarrow$ Isochoric ($V=C$)



Polytropic process

$$(q_{in} - q_{out}) + \left(\frac{W_{out} - W_{in}}{1} \right) = \Delta U$$

$$\begin{cases} \rightarrow = 0 \text{ Isentropic} \\ \rightarrow \neq 0 \text{ Polytropic} \end{cases} \quad PV^n = C \Rightarrow$$

$$W = \int_{in}^{final} P dV = \int_{in}^{final} \frac{C}{V^n} dV \quad \left| P = \frac{C}{V^n} \right.$$

$$= C \int_{in}^f \frac{1}{V^n} dV$$

$$= \frac{1}{n-1} [P_f V_f - P_i V_i]$$

$$= \frac{1}{n-1} \left(\frac{RT_f}{V_f} - \frac{RT_i}{V_i} \right) \quad \left| \begin{array}{l} PV = RT \\ P = \frac{RT}{V} \end{array} \right.$$

$$W = \frac{R}{n-1} [T_f - T_i]$$

$$W_{in} = W_{12} = \frac{R}{n-1} (T_2 - T_1)$$

$$W_{out} = W_{34} = \frac{R}{n-1} (T_4 - T_3)$$

$$W_{net} = W_{out} - W_{in}$$

Isentropic: $\frac{T_2}{T_1} = \left(\frac{V_1}{V_2} \right)^{\gamma-1}$

Polytropic $\frac{T_2}{T_1} = \left(\frac{V_1}{V_2} \right)^{n-1}$

$$T_2 = T_1 (r)^{n-1} : T_4 = T_3 \left(\frac{1}{r} \right)^{n-1}$$

* $\eta = \frac{W_{net}}{q_{in}} \rightarrow q_{2-3}$

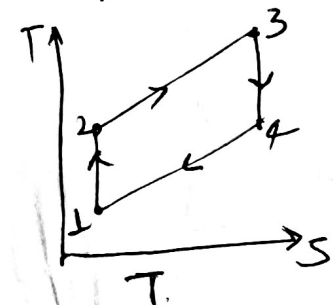
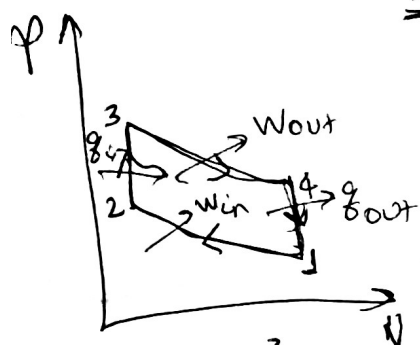
* $MEP = \frac{W_{net}}{V_1 - V_2}$

$$\Rightarrow W = C \int_i^f V^{-n} dV = C \left[\frac{V_f^{-n+1} - V_i^{-n+1}}{-n+1} \right]$$

but $PV^n = C$

$$= \frac{P_f V_f^{-n+1} - P_i V_i^{-n+1}}{-n+1}$$

$$= \frac{P_f V_f - P_i V_i}{1-n}$$



State ①
 $P_1 =$
 $T_1 =$
 $V_1 = \frac{RT_1}{P_1} =$

State 2
 $T_2 = T_1 (r)^{n-1}$
 $T_4 = T_3 \left(\frac{1}{r} \right)^{n-1}$